

HW #4 — Random Variables (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Sunday, 10/11 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as one or two .pdf files. If you do submit it as two .pdf files, submit each .pdf file as its individual submission. Any submissions not in this format are subject to point deductions.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

Background

The fourth homework will provide an introduction into random variables, particularly at discrete ones.

Textbook Problems (4 pts each)

From *Introduction to Probability, Second Edition*:

- Chapter 3: Q1, Q2, Q5, Q18, Q20

Part II: Poisson Distribution (5 pts)

Another named discrete distribution, most commonly used for waiting queues, is the Poisson distribution. The Poisson distribution allows us to find the probability of some number of the same events occurring in a certain time period. For example, a bank might be curious about how many customers might come in over a single hour or day, assuming they have some information about the rate already.

With that said, this rate is the only parameter that the Poisson distribution has, denoted as λ . This rate always refers to the items or people associated with the value of x . As such, its PMF is the following:

$$f_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \text{for } x = 0, 1, 2, \dots$$

We will investigate its properties and use R to solve probability questions involving discrete (and continuous) random variables.

1. Prove that the PMF of a Poisson distribution is a true PMF.
 - Hint: Recall the Taylor series for some summation of e^x .
2. Let $\lambda = 4$. Find the following probabilities by-hand:
 - (a) $P(X = 2; \lambda = 4)$
 - (b) $P(X = 10; \lambda = 4)$
 - (c) $P(X > 2; \lambda = 4)$
3. Write (either by-hand or typed out) the R code used to get the same probabilities seen in Q2.
4. Generate 1000 observations of $X \sim \text{Pois}(\lambda = 4)$ using R code and store this object as x . Do the same for $Y \sim \text{Pois}(\lambda = 2)$, storing the object as y . Using R, create a histogram each for x and y using the `hist()` function, describing what you see between the two distributions.
 - Screenshots of the histograms are fine here.